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# Machine learning based corrosion prediction of as cast Mg-Sn alloys for biomedical applications

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ABSTRACT

Magnesium and its various alloys are the promising materials for orthopedic implant applications because of its light weight, biocompatible and biodegradable characteristics. The prediction of corrosion rate is important when the implant material is present in the human body for longer time period and it can be done by machine learning approach. Corrosion data of Mg-Sn alloy samples were collected and fit in to the machine learning algorithm. Extreme Gradient Boost model has given the best output for the prediction of corrosion behavior with the accuracy of $R^2$ value of 0.961 in EIS plots and 0.6145 in Tafel plots.

## 1. Introduction

Magnesium is one of the biomaterials which can be highly biodegradable that suits for the application of orthopedic implant. The high corrosion rate of the implant results in faster degradation before the bone fracture gets healed. The material subjected to corrosion when it's implanted inside the body that could also support the bone healing process [1–3]. Mg and its different alloys have similar elastic modulus close to human bones which significantly reduces the stress shielding effect that are appealing for implant use [4–6]. The development of Mg alloy system is gained interest for analyzing its potential for the replacement of currently used commercial implants. As-cast and extruded Mg-Zn-Mn-(0, 1.5)Sr alloys was investigated for their corrosion behavior and cytocompatibility and it was found that Sr greatly reduces the corrosion rate and improves the cytocompatibility [7]. The bio corrosion behavior of Mg-Sn-Mn-Zn alloys were investigated subjected to various processing techniques and it was observed that extruded alloy shows least corrosion rate of 0.95 mm/y compared to cast and rolled conditions [8]. Xue etal. developed Mg-5Sn-1In alloy and observed that cathodic phase of $Mg_2Sn$ was reduced significantly owing to the addition of indium thus improves the corrosion resistance [9]. Zhao et al. stated that the eutectic $Mg_2Sn$ phase appeared as the brighter region and the greyish bands show the enrichment of Sn along the grain boundaries in Mg-5 wt%Sn alloy [10]. Recently Sn has been explored as the best option for alloying with Mg owing to its castability and corrosion resistance characteristics [11]. The addition of Sn in Mg helps to arrest the inhibition of the ions from electrolytic solution by forming $SnO_2$ in the sunbstrate [12,13]. The author in her previous study explored the corrosion inhibiting effect of Sn by adding with Mg and found that increase in addition of Sn to 5 wt% shows better corrosion resistance under various time periods [14–16]. Machine learning (ML) has attracted greater attention to predict the performance characteristics of any material and the corrosion rate of carbon steel was analyzed using random forest algorithm which produced the best results with mean square errors ranging from 0.005 to 0.093 [17]. Boxin Wei etal. used gradient boost decision tree method to model the electrochemical impedance spectroscopy (EIS) corrosion data to predict the corrosion of high entropy alloys and resulted in mean squared error of 0.9491 [18]. ML and statistical techniques were used to forecast how the addition of components and other conditions will affect the corrosion behavior of low-alloy steels in the marine environment [19]. The corrosion data of additively manufactured Inconel 718 were collected and the prediction of corrosion rate were determined by using extreme gradient boost algorithm with $R^2$ value of 0.954 in potentiodynamic polarization test and 0.997 in EIS testing [20]. It is clear from the aforementioned literatures that no research has been done on the use of machine learning for the prediction of biocorrosion of magnesium alloys in simulated body fluid (SBF). The current study thus focuses on the prediction of biocorrosion properties in Mg-xSn (x = 1,5,10 wt%) alloys using data sets acquired from Tafel and EIS tests under various immersion time period by employing machine learning approach.

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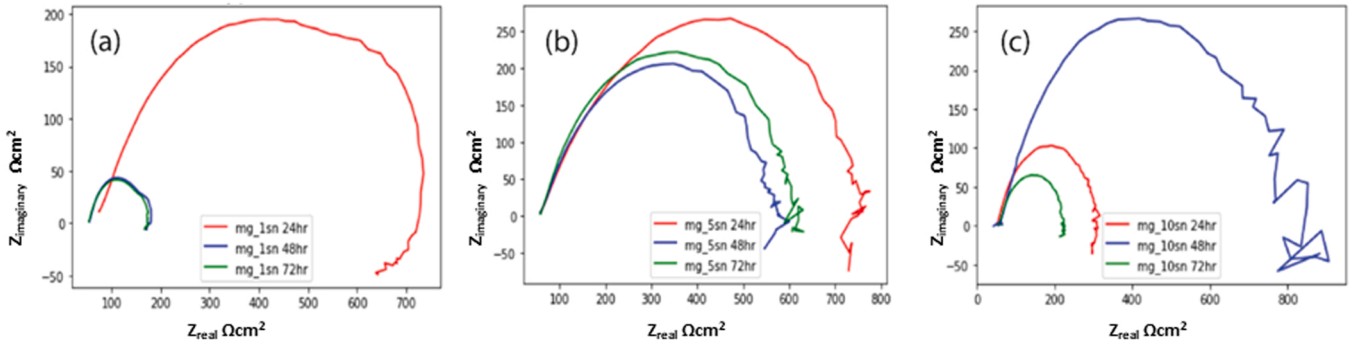



**Fig. 1.** Nyquist results for (i) Mg-1Sn (ii) Mg-5Sn (iii) Mg-10Sn for different immersion periods.

## 2. Materials and methods

Pure Mg (99.9%) and Sn (99.9%) ingots were used for the preparation of Mg-Sn alloy by casting process. Mg and different amount of Sn with nominal composition of 1 wt%, 5 wt% and 10 wt% were added together to fabricate Mg-Sn alloys by stir and squeeze casting process. The melting is carried out in a furnace at a heating temperature of 700 °C and homogeneity of mixture was enhanced by stirring at a speed of 300 rpm by the stainless stirrer. Mg-Sn melt was poured in the mold at a temperature of 750°C and it was squeezed under a pressure of 100 MPa. This provides the rapid solidification of the melt and results in cast of minimum porosity. Mg-xSn(1,5&10 wt%) alloys were cut and machined to dimensions of 10 mm in diameter and 10 mm in thickness for carrying out the corrosion experiments. The samples were polished to a grit size of 1200 before being cleaned with ethanol in order to guarantee that the surface was homogenous for the electrochemical corrosion test. The Saturated Calomel Electrode serves as the reference electrode in the three-electrode corrosion cell. Platinum wire serves as the counter electrode and the sample serves as the working electrode, respectively. Mg-Sn alloys were subjected to corrosion testing by immersing test specimens in simulated body fluid (SBF) for varying lengths of time. Using a potentiostat (Inter-face 1010, Gamry Instruments) and the sample of 0.375 $cm^2$ area exposed to SBF solution and the potentiodynamic polarization (PD) and electrochemical impedance spectroscopy (EIS) experiments were performed for Mg-Sn alloys immersed under various time period of 24, 48, and 72 h at 37 °C. The SBF was prepared by adding the chemicals in order NaCl (8.035 g), $NaHCO_3$(0.355 g), KCl (0.225 g), $K_2HPO_4$0.3 $H_2O$ (0.231 g), $MgCl_2$0.6 $H_2O$ (0.311 g), $CaCl_2$(0.292 g), $Na_2SO_4$(0.072 g), Tris (6.118 g), aM HCl(0–5 ML) to attain a pH of 7.45 [(Kokubo and Takadama 2006)]. By polarizing the sample roughly 0.3 V cathodically and anodically with reference to Open Circuit Potential at a scan rate of 0.5 mV/s after a delay of 60 min, potentiodynamic polarization graphs were produced. With an AC signal amplitude of 10 mV rms, EIS measurements were made from a start frequency of $10^4$ Hz up to 0.1 Hz. Following the tests, curve fitting and equivalent circuit modeling were performed on the Tafel and impedance data using the Gamry Echem Analyst.

## 3. Data description

Mg-Sn alloys were subjected to electrochemical corrosion experiments and the total of 100 data points were obtained for Nyquist and Bode, and 5400 data points were obtained for Potentiodynamic (PD) polarization. All these data points were loaded as a dataset for the machine learning model. The corrosion behavior of Mg-Sn alloys with different composition of Sn under various immersion time was observed using Extreme Gradient Boost, which segregated according to different plots such as Nyquist, PD and Bode plots that were obtained through the data points. Through these data points' plots, we achieved outputs, which were ordered and combined to create dataset for each condition. On these datasets, the models were performed to gain the prediction of the corrosion behavior of Mg alloy samples. The experimental results are plotted below using Nyquist, bode and potentiodynamic graphs demonstrated by Figs. 1, 2 & 3 respectively.

## 4. Model building process

With the ultimate goal to attain the prediction of the corrosion behavior, the model building process as shown in Fig. 4 initiated with experimental data collection and acquisition over the number of data points obtained from the electrochemical corrosion tests. Following this, the data was then cleaned for the further modeling processes and was split into train, validation and test datasets. After this the optimized model was developed. This model was then applied over the experiment dataset and the corrosion behavior predicted outcome was attained. The outcomes were then studied and analyzed to reach to conclusions.

## 5. Model development

Machine Learning was the building block of the model, through

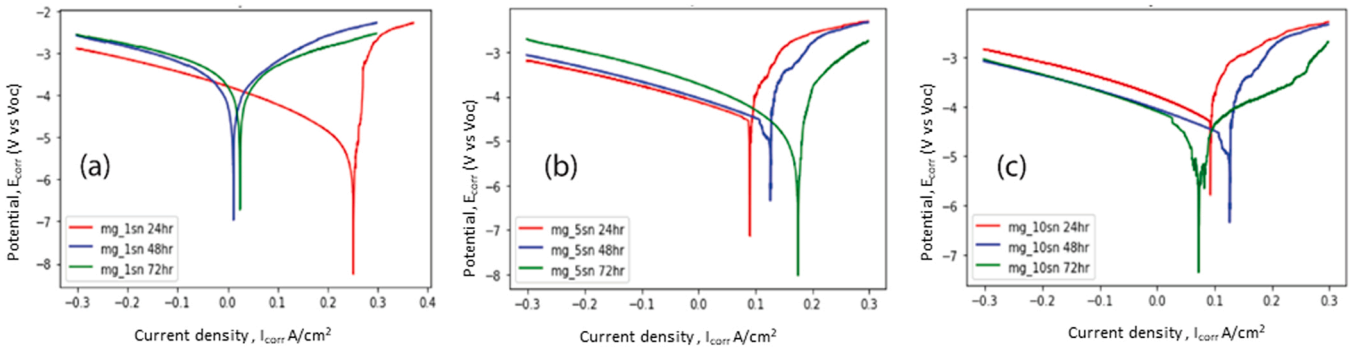



**Fig. 2.** Potentiodynamic polarization results for (i) Mg-1Sn (ii) Mg-5Sn (iii) Mg-10Sn for different immersion periods.

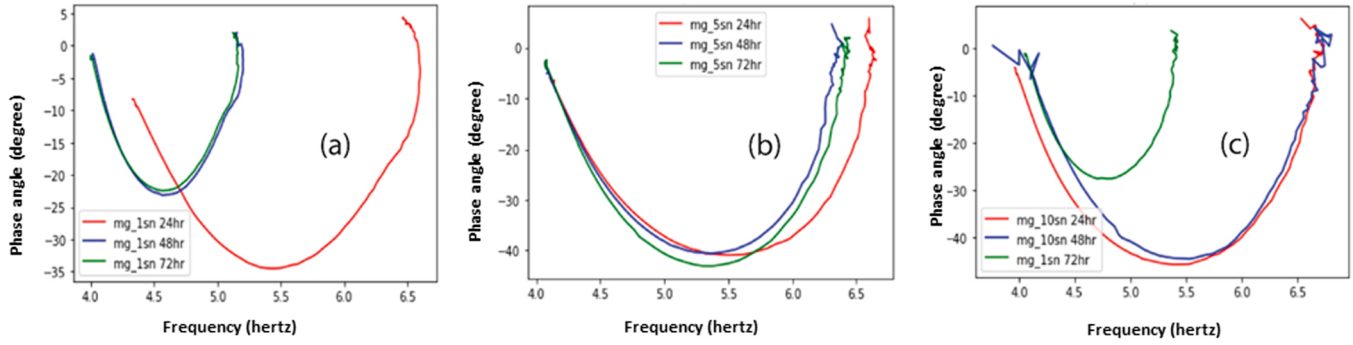



**Fig. 3.** Bode results for (i) Mg-1Sn (ii) Mg-5Sn (iii) Mg-10Sn for different immersion periods.

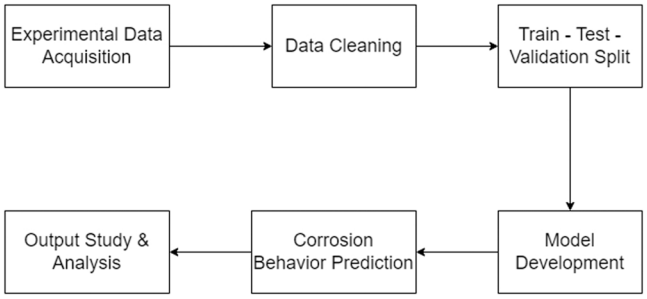



**Fig. 4.** Model building process.

which the resultant predictions were made possible. Python was used for implementation of the code and various py libraries like the Scikit-learn library was utilized to adopt the algorithm. The number of data points were used in a data set was reasonable to apply in ML model. The test, train and validation data were split in the following manner: 70%, 15% and 15% respectively. Extreme Gradient Boosting algorithm was employed to get the better results.

### 5.1. Extreme gradient boosting

Extreme Gradient Boosting, also called XGBoost, is implementation of gradient boosted decision trees. In this approach, decision trees are constructed in a sequential fashion, and all variables are then given weights before being put into the decision tree for the prediction of output. The second decision tree is then supplied with the heavier weights of the variables that were incorrectly predicted. This generates an ensemble to provide a model that is ever-more-accurate.

The objective function is to minimize the loss function and it is presented by the following expression as:

$$L(t) = \sum_i \ell(y_i, p_i(t\text{-}1) + f_t(x_i)) + \Omega(f_t) \quad (1)$$

Here, l = loss function denotes how best is the model in making predictions with the actual data. It indicates the difference between the values that are actual and predicted and $y_i$ is the true value (label) known from the training data set. Ω (ft) (regularization term) which helps to prevent overfitting of data is given below,

$$\Omega(ft) = \gamma T + (0{\cdot}5)((\lambda)(w\ 2)) \quad (2)$$

In this, T = number of trees, γ = pruning index, w = weight of the leaf, and λ = scaling factor of weights.

The objective function must be transformed to the Euclidean domain in order for us to employ conventional optimization techniques, thus we then use Taylor approximation.

Through the second-order Taylor approximation, we attain:

$$L(t) \sim \sum_I [\ell(y_i, p_i(t\text{-}1) + g_i f_t(x_i) + \tfrac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (3)$$

On removing the constant part, the simplified objective function is:

$$L(t) = \sum_i [\ g_i f_t(x_i) + \tfrac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (4)$$

After that, each tree's g and h values are computed, placed on the leaf that corresponds to them, and then combined to produce the best model possible.

### 5.2. Model validation

The model is created using XGBoost (XGB) for getting highest accuracy. The results after prediction are plotted against actual experimental data to check the model deviation. By examining similarity of curves produced after plotting, model is verified. Model is more accurate when curves are closer to each other. Initially, four different machine learning approaches were considered, namely Polynomial regression (PR), support vector machine (SVM), decision tree (DT) and extreme gradient boost (XGB). It was noticed that XGB model had the best curve fitting to the experimental data. However, the other models were not able to fit properly. This is because they were not able to detect the data regions wherein the corrosion took place. A smooth polynomial function was used to forecast the points in the PR model, resulting in a linear graph as opposed to the experimental graph, which was of no use at all. Similar to the data of experiment, the prediction curve in the SVM model is rather a gradual dip as opposed to the sharp dip of the experimental data since the kernel employed in the SVM model is a consistent and frictionless input feature space transformation. Although DT model produced results marginally better than those of the XGB model, it fell short. As a result, the XGB model was employed since it had a nearly identical prediction curve to the experimental curve and could successfully forecast the data shift area. This is due to the regularization term, also known as error reduction, which takes place at every stage of model construction. The algorithm "learns" the mistake or departure from prior right and fixes the next phase. Additionally, the algorithm offers node-optimizing parametric tuning and the depth of tree in tandem with tree development. A number of well-established measures, including $R^2$, RMSE (root mean square error) and MAE (mean absolute error) are used to assess the model's correctness. It is evident from the potentiodynamic polarization tests that the XGB model has the greatest $R^2$ at 0.954. It stands out as being superior to the other algorithms. Lower values of the measures MAE and RMSE are considered to be acceptable. The model is hyperparameter tweaked for improved output. Hyperparameters are a group of parameters that change how the machine learning model learns before it is applied to a dataset. To achieve a more effective machine learning model, which in this case tries to predict accurate corrosion behavior near to the experimental data, tuning and improving these hyperparameters is crucial.

The hyperparameter optimization equation is:

$$x^* = arg\ min_{(x\ \varepsilon\ X)} f(x) \quad (5)$$

Where f(x) represents the score of the objective to minimize the

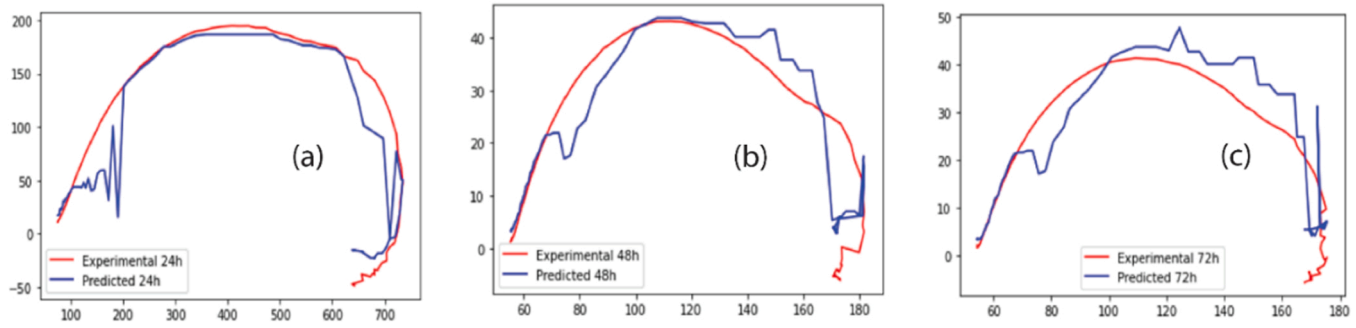



**Fig. 5.** Nyquist results for Mg-1Sn when immersed for (i) 24 h (ii) 48 h (iii).

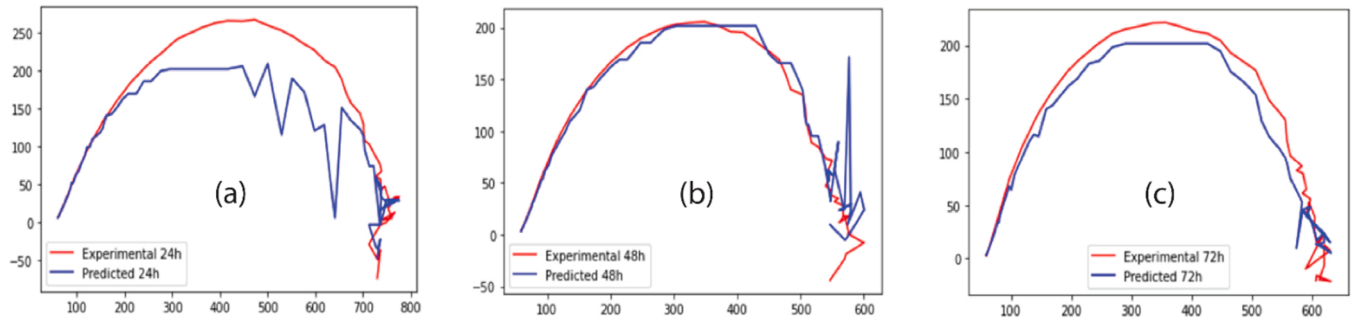



**Fig. 6.** Nyquist results for Mg-5Sn when immersed for (i) 24 h (ii) 48 h (iii) 72 h.

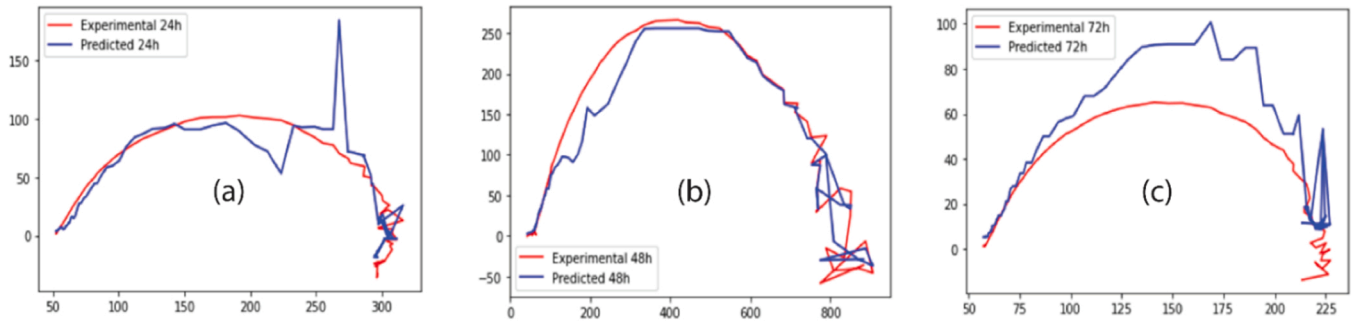



**Fig. 7.** Nyquist results for Mg-10Sn when immersed for (i) 24 h (ii) 48 h (iii) 72 h.

validation set tested error and x * represents the least scored hyper-parameter set.

For tuning the hyperparameters better, Bayesian hyperparameter optimization is used, which uses the objective function to build a probability model and utilizes that to pick the best set of hyperparameters.

According to Bayes rule,

$$p(O|h) = p(h|O) * (p(O)/p(h)) \quad (6)$$

Wherein, $p\ (O|h)$ gives the hyperparameter probability based on the objective function score. It is demonstrated by:

$$p(h|O) = l(h)\ if\ O<O* \quad (7)$$

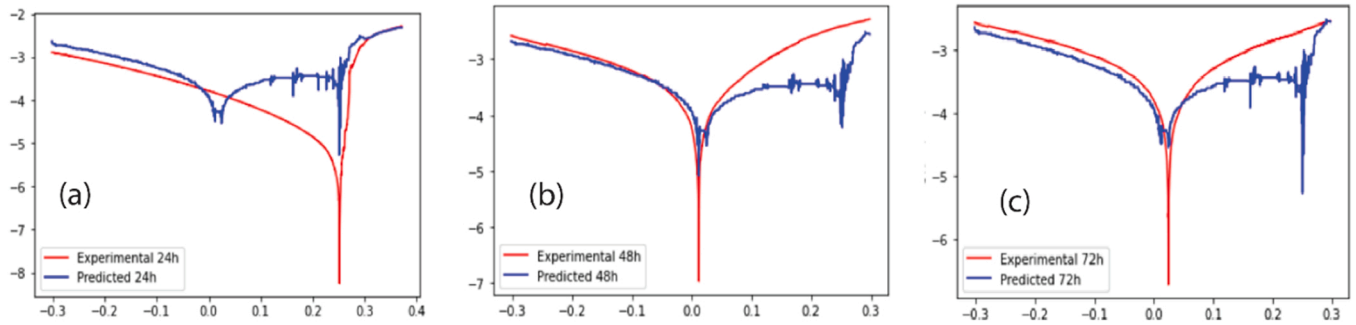



**Fig. 8.** PD results for Mg-1Sn when immersed for (i) 24 h (ii) 48 h (iii) 72 h.

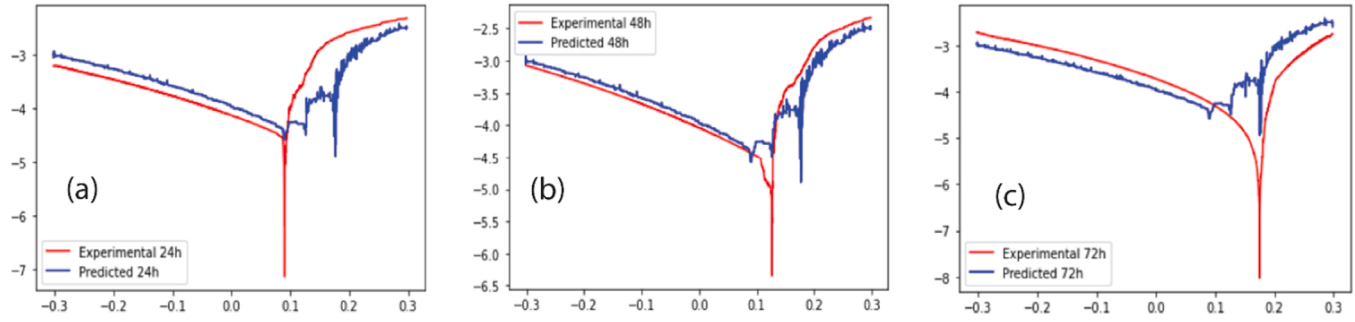



**Fig. 9.** PD results for Mg-5Sn when immersed for(i) 24 h (ii) 48 h (iii) 72 h.

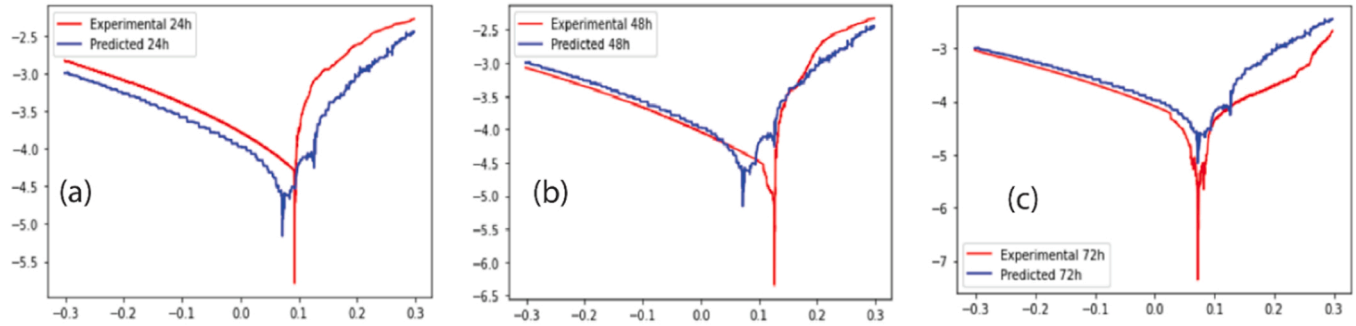



**Fig. 10.** PD results for Mg-10Sn when immersed for (i) 24 h (ii) 48 h (iii) 72 h.

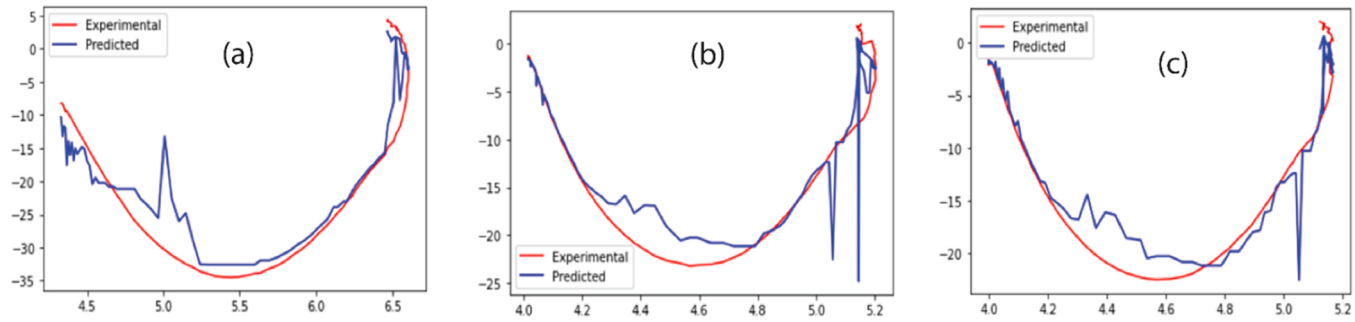



**Fig. 11.** Bode results for Mg-1Sn when immersed for (i) 24 h (ii) 48 h (iii) 72 h.

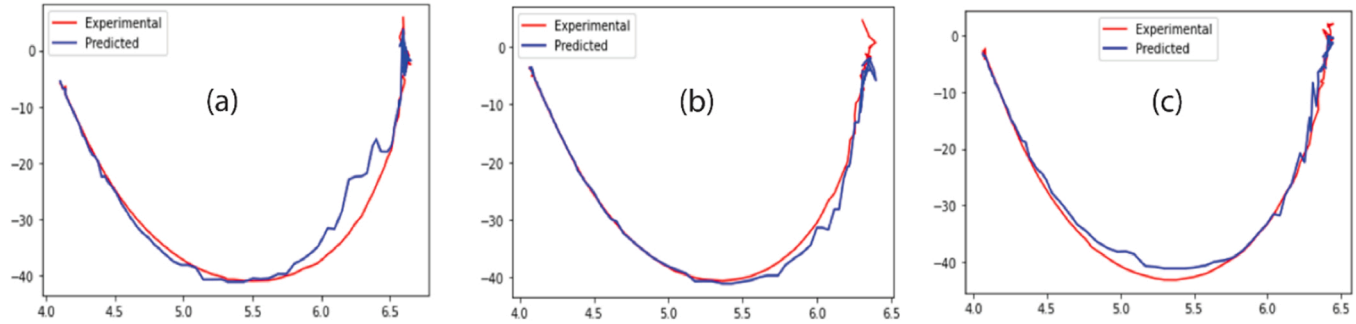



**Fig. 12.** Bode results for Mg-5Sn when immersed for (i) 24 h (ii) 48 h (iii) 72 h.

$$g(h) \text{ if } O \geq O^* \tag{8}$$

On the nyquist, PD, and bode plots, XGBoost obtained the result with the lowest RMSE.

The predictions of corrosion performance of Mg-Sn alloys with 1%, 5% and 10 wt% tin are analyzed for 3 different time periods i.e., 24 h, 48 h and 72 h respectively. This helps to determine the optimal composition of Mg-Sn alloy for better corrosion resistance. The points are plotted as bode, nyquist and PD to check the trends followed by these materials. It is observed that Mg-Sn (1%, 5% and 10%) for 48 hr immersion gave the highest accuracy, of about 96.19% for bode plot.

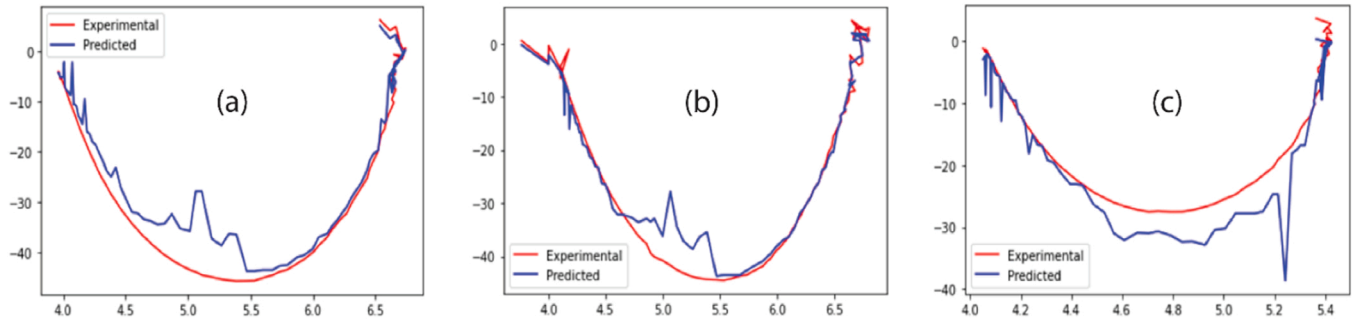



**Fig. 13.** Bode results for Mg-10Sn when immersed for (i) 24 h (ii) 48 h (iii) 72 h.

Accuracy for Nyquist and PD were slightly less but still better than other ML algorithms. The models are grouped according to the percentage of tin used. Each group has results for 24 h, 48 h and 72 h of immersion. The Figs. 5, 6 and 7 describe the Nyquist plot behavior, Figs. 8, 9 and 10 describe the potentiodynamic results and Figs. 11, 12 and 13 depict the Bode results for Mg-xSn(x = 1, 5, 10) for different immersion time.

Extreme gradient boost machine learning model has given better prediction of corrosion performance of the Mg-Sn alloys under various immersion time. The results are highly acceptable with the experimental results and it can be further used for long term immersion. This intend to analyze the corrosion behavior of material in SBF when it can be used as an implant for treating bone fractures. Mg-Sn alloy with 5 wt% tin shows the better corrosion resistance compared to other alloy system. The formation of strong protective oxide layer ($SnO_2$ +$Mg(OH)_2$) and also the apatite growth on the surface prevents the material from further corrosion. ML greatly helps to understand the mechanism of corrosion by the predicted results which helps to reduce the experimentation time and also significantly contributes for understanding the material performance for biomedical applications.

## 6. Conclusion

In this study, machine learning approaches were utilized to forecast the corrosion behaiour of Mg-xSn(1,5 and 10 wt%) alloys when its immersed in SBF for 24, 48, and 72 h. Data from experiments were integrated in the XGB algorithm. Utilizing experimental data, a machine learning strategy was used to investigate the connection between post-processing methods and corrosion resistance. Extreme gradient boosting was used to show that this strategy is feasible. The Mg-5Sn corrosion rate was predicted by the ML-based model with the best degree of accuracy using the XGB technique, with an $R^2$ value of 0.961 in the bode plot, 0.87 for Nquist and 0.85 for PD plots. Mg-Sn alloy samples immersed in SBF for varying lengths of time have shown that the XGB algorithm has a higher degree of accuracy in predicting corrosion rates.

## Declaration of Competing Interest

The authors declare there is no conflict of interest.

## Data availability

Data will be made available on request.

## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.mtcomm.2023.106108.

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